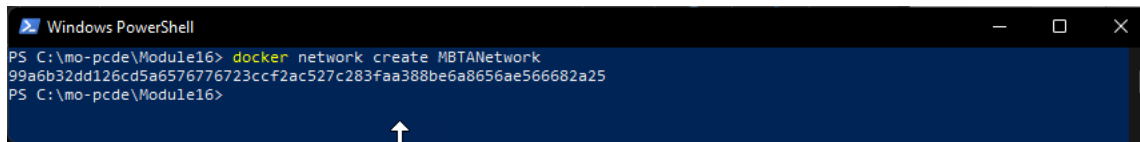


Project 16.1 Solution File

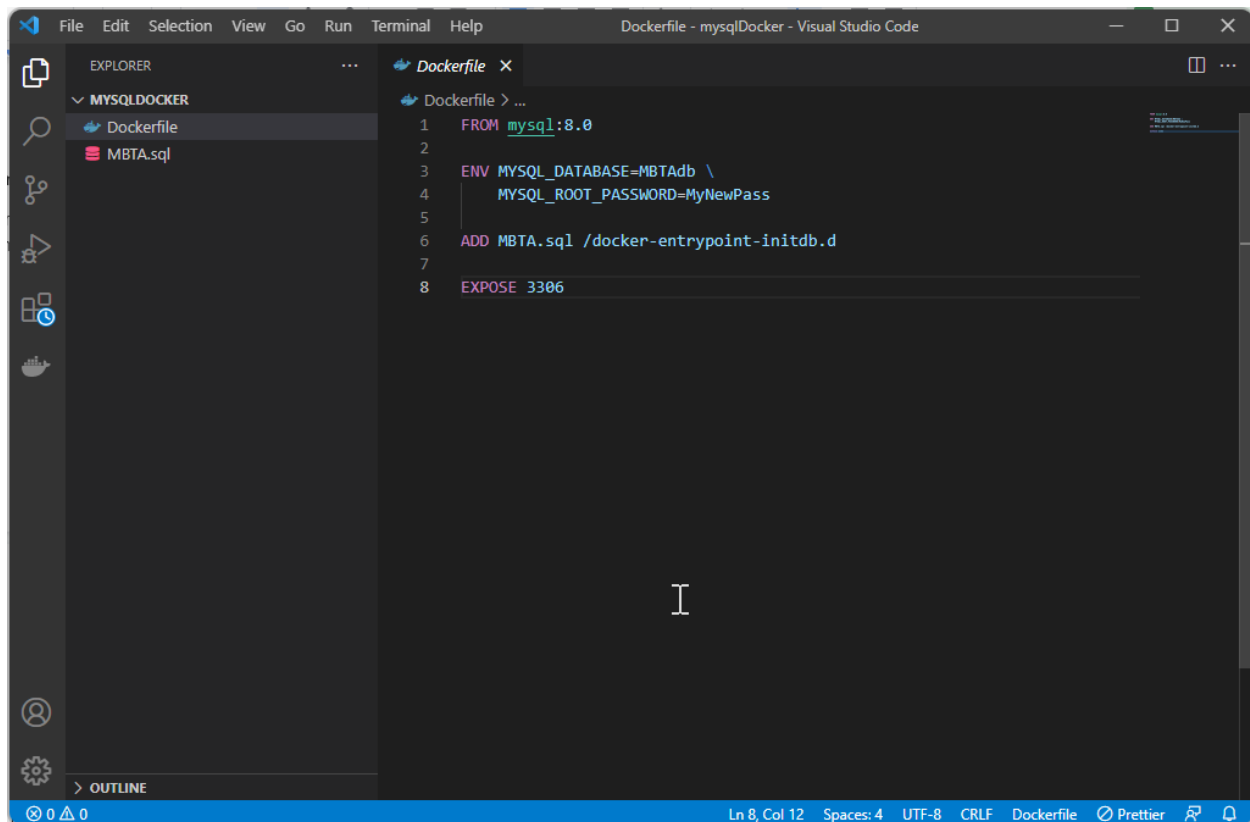
1.



```
Windows PowerShell
PS C:\mo-pcde\Module16> docker network create MBTANetwork
99a6b32dd126cd5a6576776723ccf2ac527c283faa388be6a8656ae566682a25
PS C:\mo-pcde\Module16>
```

An upward-pointing arrow is located below the terminal window.

2.



```
File Edit Selection View Go Run Terminal Help Dockerfile - mysqlDocker - Visual Studio Code
EXPLORER
  MySQLDOCKER
    Dockerfile
    MBTA.sql
  Dockerfile
  Dockerfile > ...
  1 FROM mysql:8.0
  2
  3 ENV MYSQL_DATABASE=MBTadb \
  4     MYSQL_ROOT_PASSWORD=MyNewPass
  5
  6 ADD MBTA.sql /docker-entrypoint-initdb.d
  7
  8 EXPOSE 3306
  Ln 8, Col 12 Spaces: 4 UTF-8 CRLF Dockerfile Prettier
```

The screenshot shows the Visual Studio Code interface with a Dockerfile open. The Explorer sidebar on the left shows a project named 'MYSQLDOCKER' containing 'Dockerfile' and 'MBTA.sql'. The Dockerfile content is as follows:

```
1 FROM mysql:8.0
2
3 ENV MYSQL_DATABASE=MBTadb \
4     MYSQL_ROOT_PASSWORD=MyNewPass
5
6 ADD MBTA.sql /docker-entrypoint-initdb.d
7
8 EXPOSE 3306
```

The status bar at the bottom indicates the file is 'Ln 8, Col 12', uses 'Spaces: 4', 'UTF-8' encoding, 'CRLF' line endings, and is formatted with 'Prettier'.

a)

```

1 CREATE DATABASE IF NOT EXISTS MBTAdb;
2
3 USE MBTAdb;
4
5 DROP TABLE IF EXISTS mbta_buses;
6
7 CREATE TABLE mbta_buses (
8     record_num INT AUTO_INCREMENT PRIMARY KEY,
9     id varchar(255) not null,
10    latitude decimal(11,8) not null,
11    longitude decimal(11,8) not null
12 );
13
14

```

b)

```

In [1]: mbtaURL = "https://api-v3.mbta.com/vehicles?filter[route]=1&include=trip"

import urllib.request, json
with urllib.request.urlopen(mbtaURL) as url:
    data = json.loads(url.read().decode())

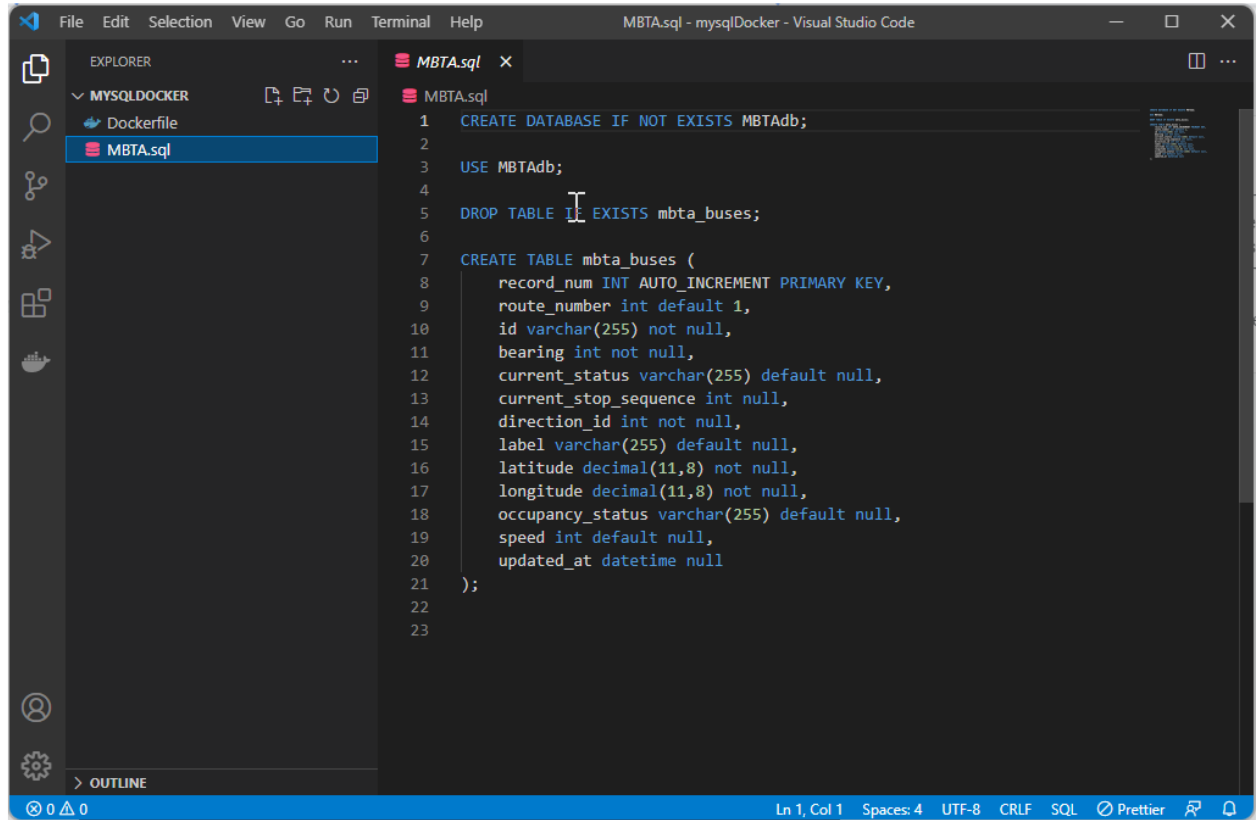
    with open('data.json', 'w') as outfile:
        json.dump(data, outfile)

    with open('data.txt', 'w') as outfile:
        json.dump(json.dumps(data, indent=4, sort_keys=True), outfile)

    print(json.dumps(data, indent=4, sort_keys=True))

{
  "data": [
    {
      "attributes": {
        "bearing": 0,
        "current_status": "IN_TRANSIT_TO",
        "current_stop_sequence": null,
        "direction_id": 0,
        "label": "1843",
        "latitude": 42.34122608,
        "longitude": -71.050801,
        "occupancy_status": null,
        "speed": null,
        "updated_at": "2021-12-03T02:24:16-05:00"
      },
      "id": "y1843",
      "links": {
        "self": "/vehicles/y1843"
      },
      "relationships": {
        "route": {
          "data": {
            "id": "1",
            "type": "route"
          }
        },
        "stop": {
          "data": null
        },
        "trip": {
          "data": {
            "id": "1"
          }
        }
      }
    }
  ]
}

```

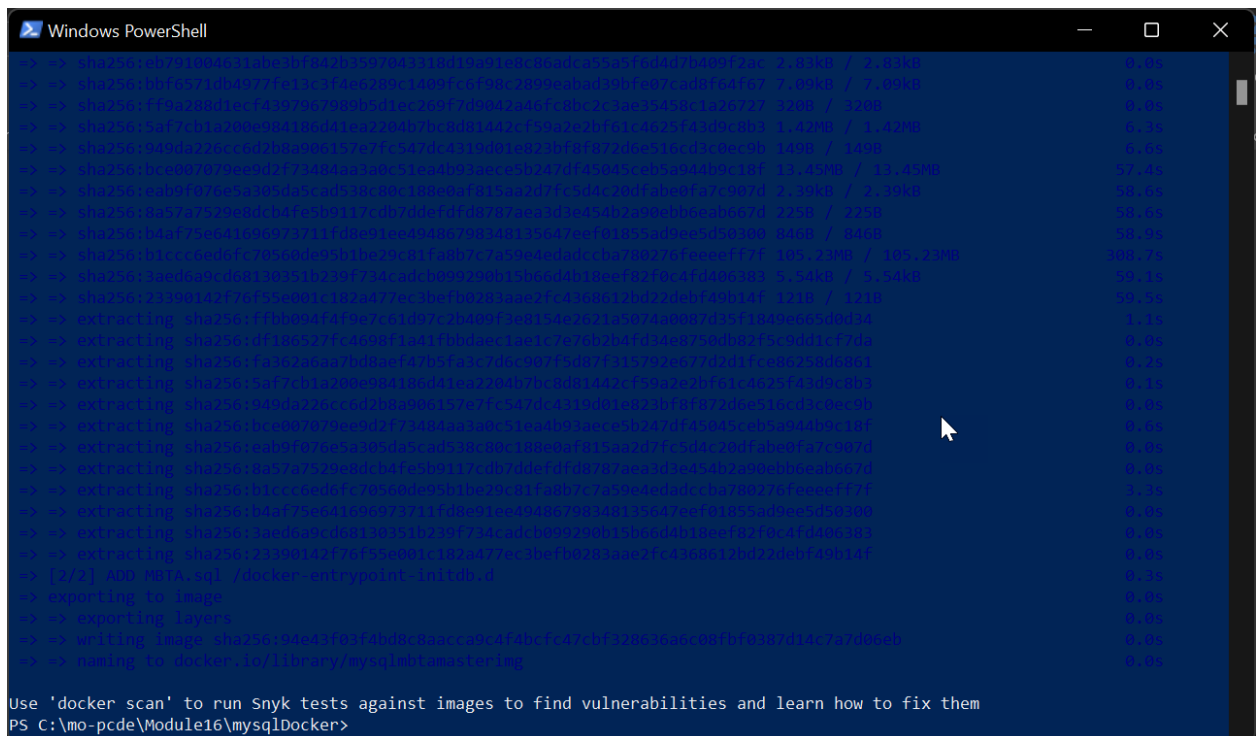


```

1 CREATE DATABASE IF NOT EXISTS MBTAdb;
2
3 USE MBTAdb;
4
5 DROP TABLE IF EXISTS mbta_buses;
6
7 CREATE TABLE mbta_buses (
8     record_num INT AUTO_INCREMENT PRIMARY KEY,
9     route_number int default 1,
10    id varchar(255) not null,
11    bearing int not null,
12    current_status varchar(255) default null,
13    current_stop_sequence int null,
14    direction_id int not null,
15    label varchar(255) default null,
16    latitude decimal(11,8) not null,
17    longitude decimal(11,8) not null,
18    occupancy_status varchar(255) default null,
19    speed int default null,
20    updated_at datetime null
21 );
22
23

```

c)

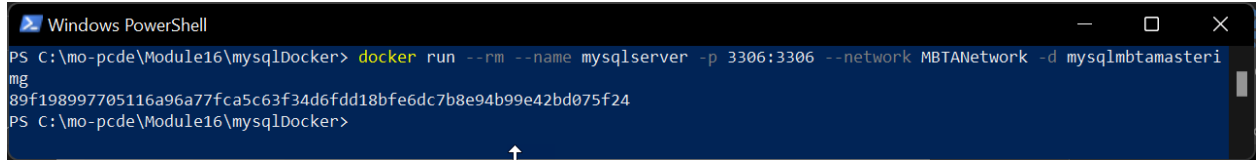


```

PS C:\> docker build -f Dockerfile -t mysql:mbta .
[+] Building 0.0s
[+] Exporting to image
[+] Exporting layers
[+] Writing image sha256:0a41f03f4b088aac0ac43bfc47c1f328636a0a8fbf03b7d14c7a7d8eab
[+] Naming to docker.io/library/mysql:mbta
PS C:\> docker image ls
REPOSITORY          TAG                 IMAGE ID            SIZE
mysql               mbta                0a41f03f4b088aac0ac43bfc47c1f328636a0a8fbf03b7d14c7a7d8eab
PS C:\> docker export --output mbta.mysql:mbta 0a41f03f4b088aac0ac43bfc47c1f328636a0a8fbf03b7d14c7a7d8eab
PS C:\>

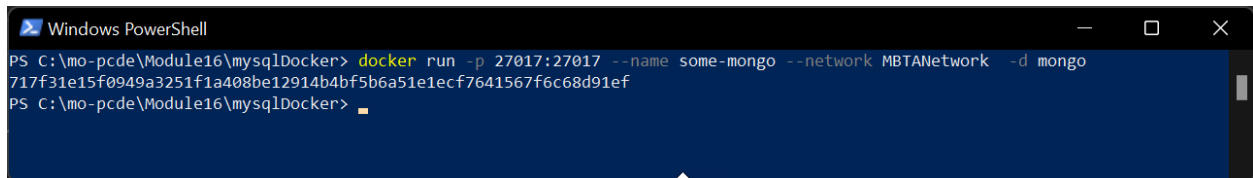
```

d)



```
Windows PowerShell
PS C:\mo-pcde\Module16\mysqlDocker> docker run --rm --name mysqlserver -p 3306:3306 --network MBTANetwork -d mysqlmbtamasteri
mg
89f198997705116a96a77fca5c63f34d6fdd18bfe6dc7b8e94b99e42bd075f24
PS C:\mo-pcde\Module16\mysqlDocker>
```

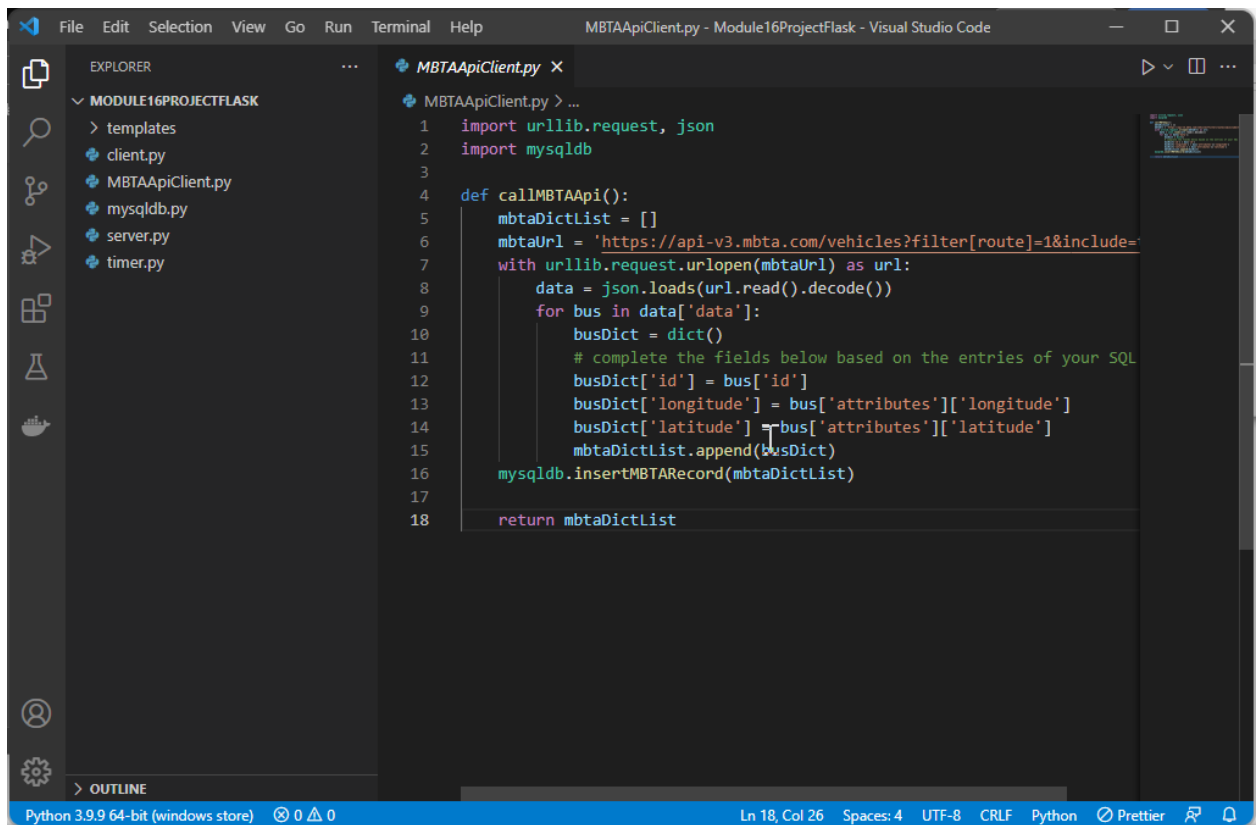
3.



```
Windows PowerShell
PS C:\mo-pcde\Module16\mysqlDocker> docker run -p 27017:27017 --name some-mongo --network MBTANetwork -d mongo
717f31e15f0949a3251f1a408be12914b4bf5b6a51e1ecf7641567f6c68d91ef
PS C:\mo-pcde\Module16\mysqlDocker>
```

4.

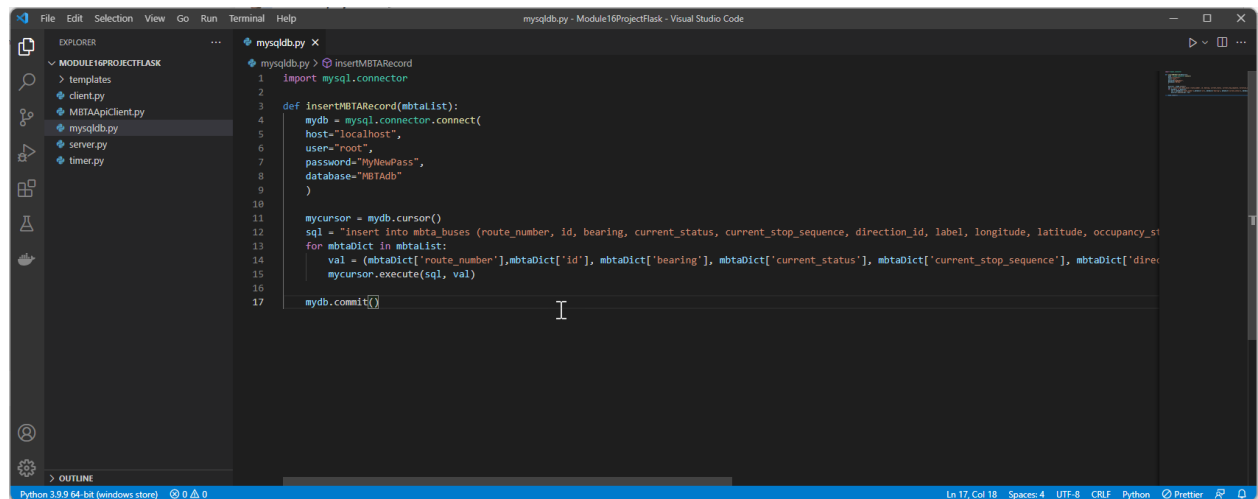
a)



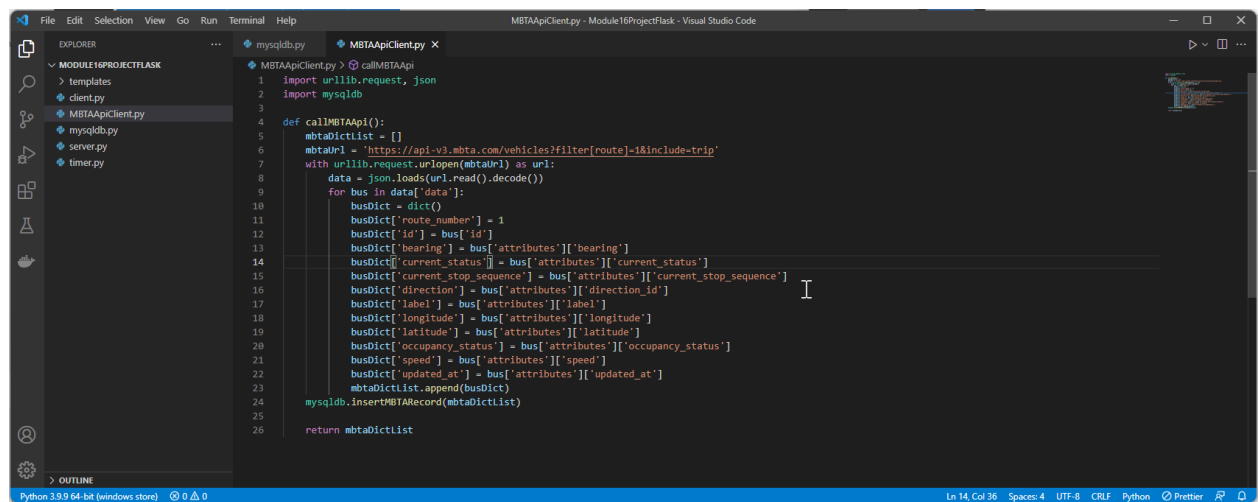
```
File Edit Selection View Go Run Terminal Help MBTAApiClient.py - Module16ProjectFlask - Visual Studio Code
EXPLORER
  MODULE16PROJECTFLASK
    > templates
    client.py
    MBTAApiClient.py
    mysqlDb.py
    server.py
    timer.py
OUTLINE
Python 3.9.9 64-bit (windows store) 0 0 0 Ln 18, Col 26 Spaces: 4 UTF-8 CRLF Python Prettier

1 import urllib.request, json
2 import mysqlDb
3
4 def callMBTAApi():
5     mbtaDictList = []
6     mbtaUrl = 'https://api-v3.mbtta.com/vehicles?filter[route]=1&include=
7     with urllib.request.urlopen(mbtaUrl) as url:
8         data = json.loads(url.read().decode())
9         for bus in data['data']:
10             busDict = dict()
11             # complete the fields below based on the entries of your SQL
12             busDict['id'] = bus['id']
13             busDict['longitude'] = bus['attributes']['longitude']
14             busDict['latitude'] = bus['attributes']['latitude']
15             mbtaDictList.append(busDict)
16     mysqlDb.insertMBTAREcord(mbtaDictList)
17
18     return mbtaDictList
```

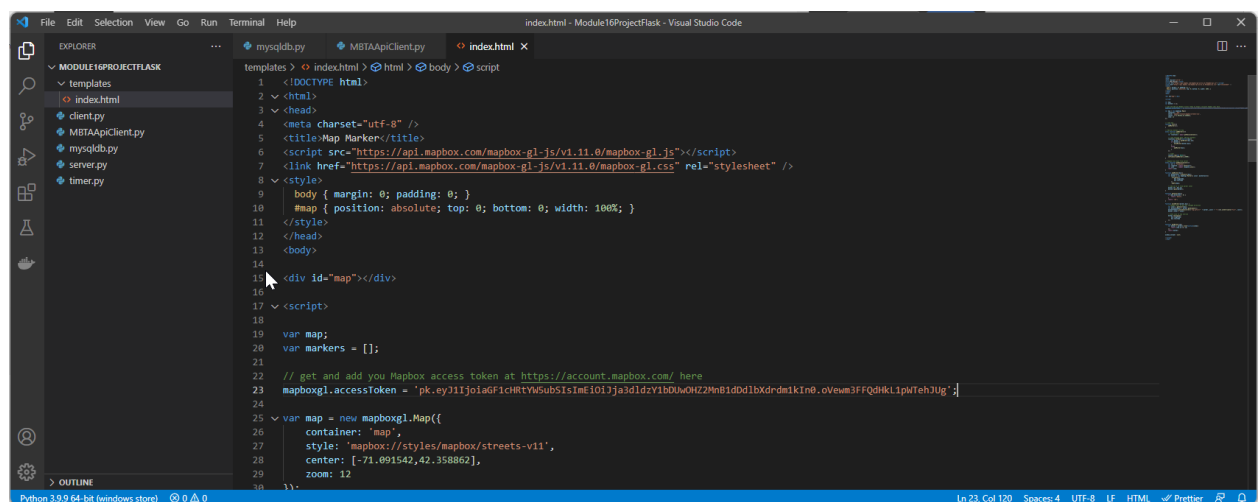
b)



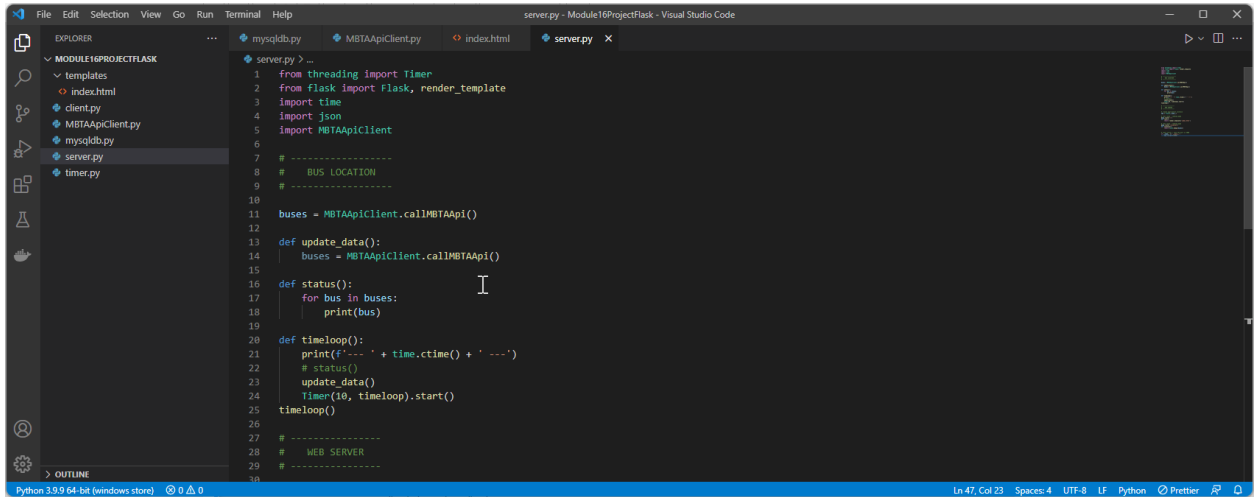
c)



d)



e)

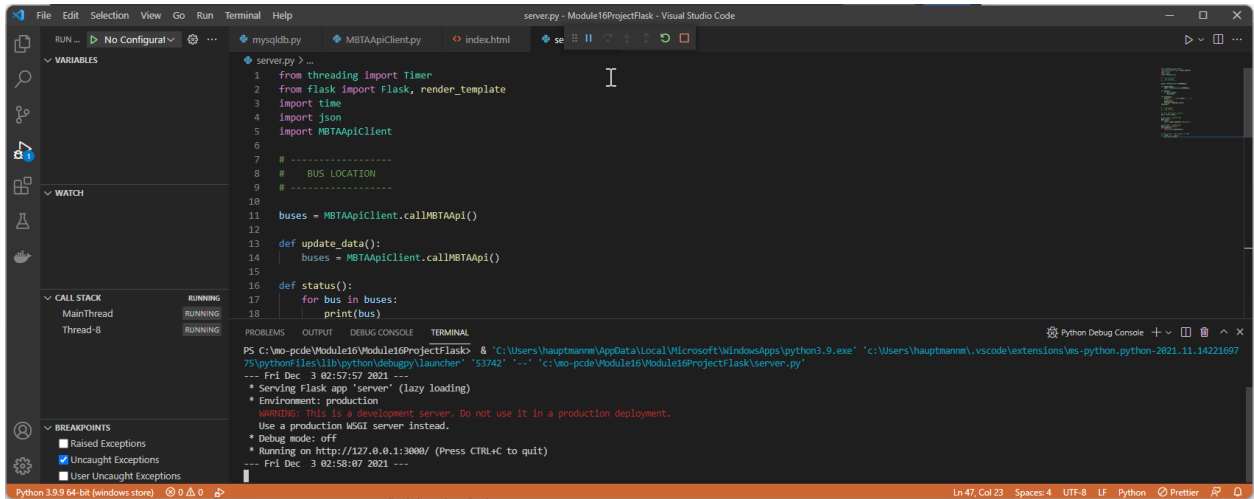


```

server.py > ...
1 from threading import Timer
2 from flask import Flask, render_template
3 import time
4 import json
5 import MBTAApiClient
6
7 # -----
8 # BUS LOCATION
9 # -----
10
11 buses = MBTAApiClient.callMBTAApi()
12
13 def update_data():
14     buses = MBTAApiClient.callMBTAApi()
15
16 def status():
17     for bus in buses:
18         print(bus)
19
20 def timeLoop():
21     print(f'--- ' + time.ctime() + ' ---')
22     # status()
23     update_data()
24     Timer(10, timeLoop).start()
25 timeLoop()
26
27 # -----
28 # WEB SERVER
29 # -----
30

```

f)



```

server.py > ...
1 from threading import Timer
2 from flask import Flask, render_template
3 import time
4 import json
5 import MBTAApiClient
6
7 # -----
8 # BUS LOCATION
9 # -----
10
11 buses = MBTAApiClient.callMBTAApi()
12
13 def update_data():
14     buses = MBTAApiClient.callMBTAApi()
15
16 def status():
17     for bus in buses:
18         print(bus)
19

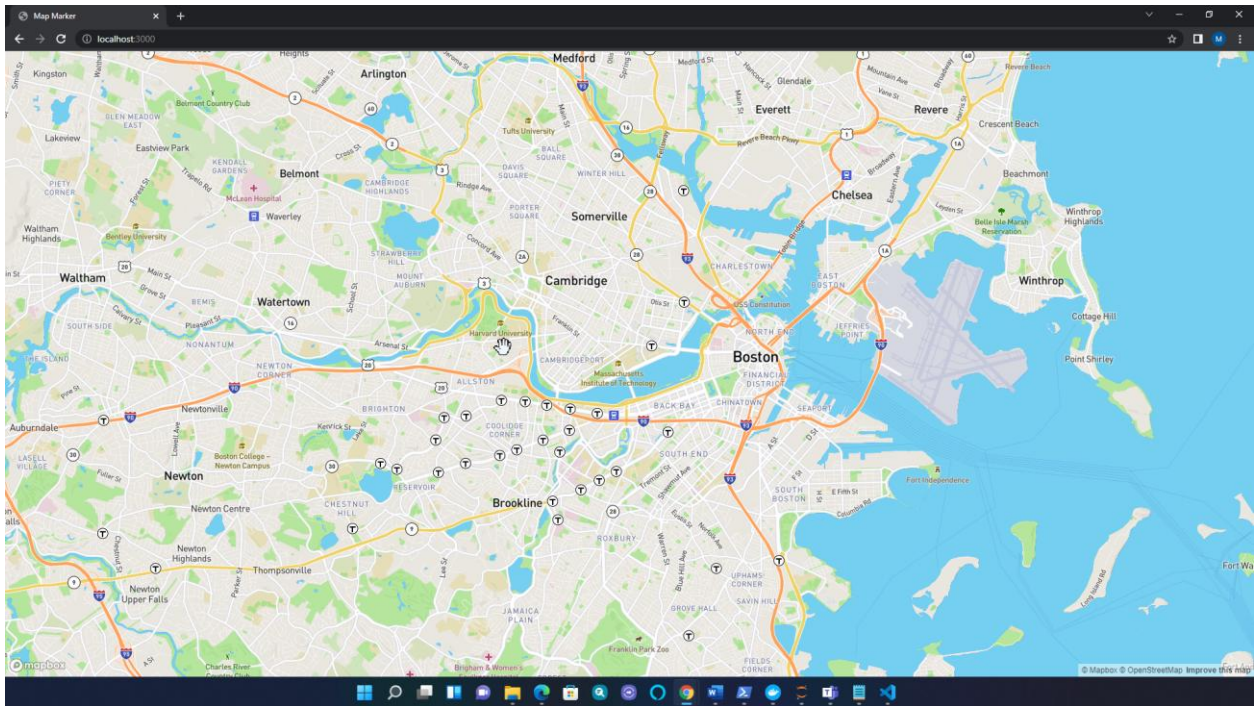
```

```

PS C:\no-pcde\Module16\ProjectFlask> & 'C:\Users\hauptmann\AppData\Local\Microsoft\WindowsApps\python3.9.exe' 'c:\Users\hauptmann\.vscode\extensions\ms-python.python-2021.11.1422169775\pythonfiles\lib\python\debugpy\launcher' '53742' '-' 'c:\no-pcde\Module16\ProjectFlask\server.py'
--- Fri Dec 3 02:57:57 2021 ---
* Serving Flask app 'server' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on http://127.0.0.1:3000/ (Press CTRL+C to quit)
--- Fri Dec 3 02:58:07 2021 ---

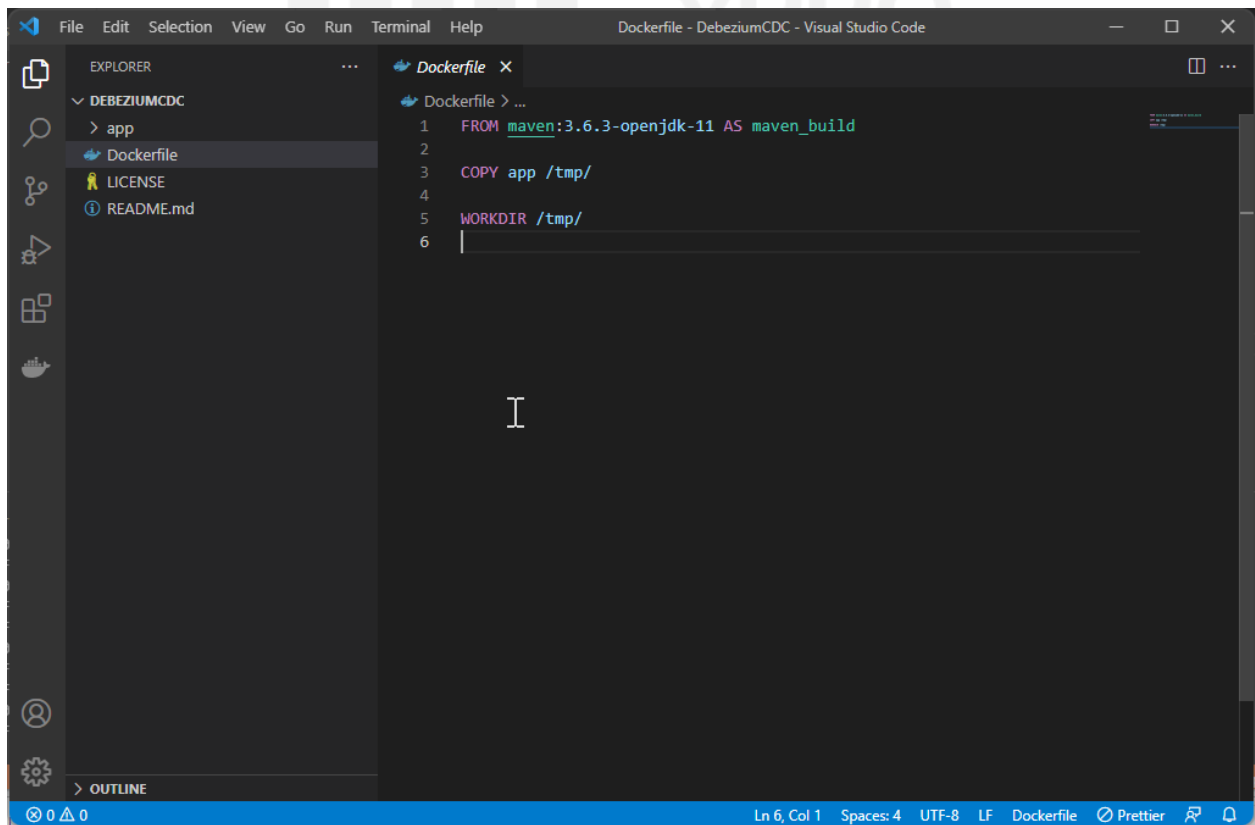
```

g)



5.

a.



b.

```
Windows PowerShell
PS C:\mo-pcde\Module16\DebeziumCDC> docker build -t debeziummodule16 .
[+] Building 0.1s (8/8) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 111B 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load metadata for docker.io/library/maven:3.6.3-openjdk-11 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 26.46kB 0.0s
=> CACHED [1/3] FROM docker.io/library/maven:3.6.3-openjdk-11 0.0s
=> [2/3] COPY app /tmp/ 0.0s
=> [3/3] WORKDIR /tmp/ 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:151f8b20e9a201fae513072be70484f51cd7dd5209f01dc731ca2fc15cc67 0.0s
=> => naming to docker.io/library/debeziummodule16 0.0s

Use 'docker scan' to run Snyk tests against images to find vulnerabilities and learn how to fix them
PS C:\mo-pcde\Module16\DebeziumCDC>
```

c.

```
Windows PowerShell
PS C:\mo-pcde\Module16\DebeziumCDC> docker run -it --rm --name debeziumserver --network MBTANetwork debeziummodule16 bash
root@1e1957fdf939:/tmp#
```

d.

```
docker exec -it 1e1957fdf939fe36312e4462466bf4f7dedeff443dcf9f18418e96355c0d0e6c /bin/sh
Get:1 http://deb.debian.org/debian buster InRelease [122 kB]
Get:2 http://security.debian.org/debian-security buster/updates InRelease [65.4 kB]
Get:3 http://security.debian.org/debian-security buster/updates/main amd64 Packages [309 kB]
Get:4 http://deb.debian.org/debian buster-updates InRelease [51.9 kB]
Get:5 http://deb.debian.org/debian buster/main amd64 Packages [7906 kB]
Get:6 http://deb.debian.org/debian buster-updates/main amd64 Packages [15.2 kB]
Fetched 8470 kB in 37s (226 kB/s)
Reading package lists... Done
root@1e1957fdf939:/tmp# apt-get install nano
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  spell
The following NEW packages will be installed:
  nano
0 upgraded, 1 newly installed, 0 to remove and 21 not upgraded.
Need to get 544 kB of archives.
After this operation, 2269 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian buster/main amd64 nano amd64 3.2-3 [544 kB]
Fetched 544 kB in 2s (288 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package nano.
(Reading database ... 12572 files and directories currently installed.)
Preparing to unpack .../archives/nano_3.2-3_amd64.deb ...
Unpacking nano (3.2-3) ...
Setting up nano (3.2-3) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
root@1e1957fdf939:/tmp#
```

e.

```

Select docker exec -it 1e1957fd939fe36312e4462466bf4f7dedeff443dcf9f18418e96355c0d0e6c /bin/sh
GNU nano 3.2 /src/main/java/mit/edu/tv/listener/MongoDB.java

String connectionString = "mongodb://some-mongo:27017";
try {
    MongoClient mongoClient = MongoClient.create(connectionString);
    List<Document> databases = mongoClient.listDatabases().into(new ArrayList<>());
    databases.forEach(db -> System.out.println(db.toJson()));
} catch (Exception e)
{
}

}

public void insertRecord(String record)
{
    String connectionString = "mongodb://some-mongo:27017";
    try {
        // Add code here
        MongoClient mongoClient = MongoClient.create(connectionString);
        MongoDB database = mongoClient.getDatabase("myDatabase");
        Document document = new Document();
        document.append("recordId", "CDC");
        document.append("value", record);
        database.getCollection("myCollection").insertOne(document);
    } catch (Exception e)
    {
    }
}

File to insert [from .]:
^G Get Help      M-F New Buffer    ^X Execute Command
^C Cancel        M-N No Conversion ^T To Files

```

f.

```

docker exec -it 1e1957fd939fe36312e4462466bf4f7dedeff443dcf9f18418e96355c0d0e6c /bin/sh
GNU nano 3.2 /tmp/src/main/java/mit/edu/tv/listener/DebeziumListener.java

this.debeziumEngine = DebeziumEngine.create(ChangeEventFormat.of(Connect.class))
    .using(customerConnectorConfiguration.asProperties())
    .notifying(this::handleChangeEvent)
    .build();

// this.customerService = customerService;
}

private void handleChangeEvent(RecordChangeEvent<SourceRecord> sourceRecordRecordChangeEvent) {
    SourceRecord sourceRecord = sourceRecordRecordChangeEvent.record();

    MongoDB mongoDB = new MongoDB();
    mongoDB.testConnection();
    // Use the insertRecord method to insert a record inside the MongoDB database.
    // Pass sourceRecord.value().toString()
    mongoDB.insertRecord(sourceRecord.value().toString());

    System.out.println("Key = " + sourceRecord.key() + " value = " + sourceRecord.value() + "");
}

@PostConstruct
private void start() {
    this.executor.execute(debeziumEngine);
}

^G Get Help      ^O Write Out      ^W Where Is      ^K Cut Text      ^J Justify      ^C Cur Pos      M-U Undo      M-A Mark Text
^X Exit          ^R Read File      ^L Replace       ^U Uncut Text    ^T To Spell     ^_ Go To Line    M-E Redo      M-6 Copy Text

```

g.

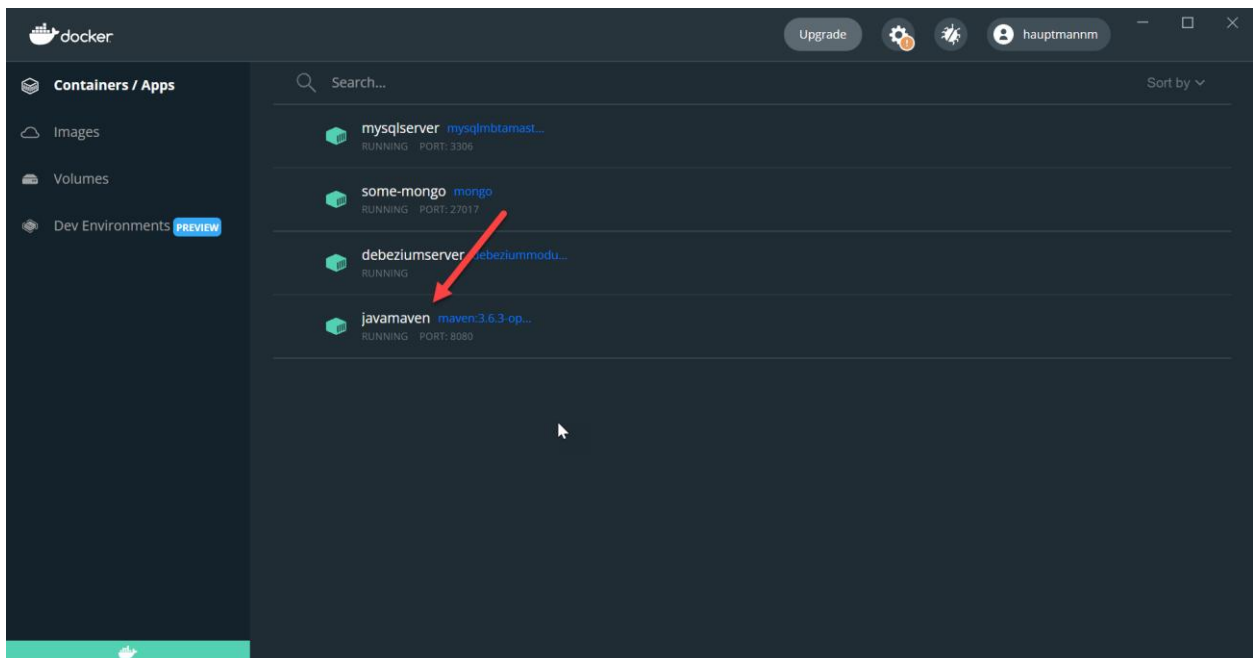
```

docker exec -it 1e1957fd939fe36312e4462466bf4f7dedeff443dc9f18418e96355c0d0e6c /bin/sh
startBinlogFilename=binlog.000002, restartBinlogPosition=156, restartRowsToSkip=0, restartEventsToSkip=0, currentEventLengthInBytes=0, inTransaction=false, transactionId=null]]
2021-12-03 08:32:18.458 INFO 586 --- [rce-coordinator] io.debezium.util.Threads : Requested thread factory for connector MySQLConnector, id = localhost_MBTADB named = binlog-client
2021-12-03 08:32:18.459 INFO 586 --- [rce-coordinator] i.d.p.ChangeEventSourceCoordinator : Starting streaming
2021-12-03 08:32:18.465 INFO 586 --- [rce-coordinator] i.d.c.m.MySqlStreamingChangeEventSource : Skip 0 events on streaming start
2021-12-03 08:32:18.465 INFO 586 --- [rce-coordinator] i.d.c.m.MySqlStreamingChangeEventSource : Skip 0 rows on streaming start
2021-12-03 08:32:18.466 INFO 586 --- [rce-coordinator] io.debezium.util.Threads : Creating thread debezium-mysqlconnector-localhost_MBTADB-binlog-client
2021-12-03 08:32:18.469 INFO 586 --- [ysqlserver:3306] io.debezium.util.Threads : Creating thread debezium-mysqlconnector-localhost_MBTADB-binlog-client
2021-12-03 08:32:18.476 INFO 586 --- [ysqlserver:3306] c.g.shyiko.mysql.binlog.BinaryLogClient : Connected to mysqlserver:3306 at binlog.000002/156 (sid:10181, cid:207)
2021-12-03 08:32:18.476 INFO 586 --- [ysqlserver:3306] i.d.c.m.MySqlStreamingChangeEventSource : Connected to MySQL binlog at mysqlserver:3306, starting at MySQLOffsetContext [sourceInfoSchema=Schema{io.debezium.connector.mysql.SourceSTRUCT}, sourceInfo=SourceInfo [currentGtid=null, currentBinlogFilename=binlog.000002, currentBinlogPosition=156, currentRowNumber=0, serverId=0, sourceTime=2021-12-03T08:32:18.433Z, threadId=-1, currentQuery=null, tableIds=[MBTADB.mbt_a_buses], databaseName=MBTADB], partition={server=localhost_MBTADB}, snapshotCompleted=true, transactionContext=TransactionContext [currentTransactionId=null, perTableEventCount={}, totalEventCount=0], restartGtidSet=null, restartBinlogFilename=binlog.000002, restartBinlogPosition=156, restartRowsToSkip=0, restartEventsToSkip=0, currentEventLengthInBytes=0, inTransaction=false, transactionId=null]
2021-12-03 08:32:18.476 INFO 586 --- [rce-coordinator] i.d.c.m.MySqlStreamingChangeEventSource : Waiting for keepalive thread to start
2021-12-03 08:32:18.477 INFO 586 --- [ysqlserver:3306] io.debezium.util.Threads : Creating thread debezium-mysqlconnector-localhost_MBTADB-binlog-client
2021-12-03 08:32:18.577 INFO 586 --- [rce-coordinator] i.d.c.m.MySqlStreamingChangeEventSource : Keepalive thread is running

```

6.

a.



b.

```
docker exec -it 4f38d2f55f7cb8e88b5fb15b28b2f41ac318e801b537bfc39f4812baf9d9b7d5 /bin/sh
root@4f38d2f55f7c:/java-quick-start-master# cd /java-quick-start-master/src/main/java/com/mongodb/quickstart
root@4f38d2f55f7c:/java-quick-start-master/src/main/java/com/mongodb/quickstart# ls -l
total 52
-rwxr-xr-x 1 root root 3131 Nov 27 10:40 AggregationFramework.java
-rwxr-xr-x 1 root root 4991 Nov 27 10:40 ChangeStreams.java
-rwxr-xr-x 1 root root 592 Nov 27 10:40 Connection.java
-rwxr-xr-x 1 root root 2396 Nov 27 10:40 Create.java
-rwxr-xr-x 1 root root 1585 Nov 27 10:40 Delete.java
-rwxr-xr-x 1 root root 161 Nov 27 10:40 HelloMongoDB.java
-rwxr-xr-x 1 root root 3170 Nov 27 10:40 MappingPOJO.java
-rwxr-xr-x 1 root root 3026 Nov 27 10:40 Read.java
-rwxr-xr-x 1 root root 868 Dec 3 10:18 ReadCDC.java
-rwxr-xr-x 1 root root 4060 Nov 27 10:40 Update.java
drwxr-xr-x 2 root root 4096 Nov 27 10:40 csfle
drwxr-xr-x 2 root root 4096 Nov 27 10:40 models
root@4f38d2f55f7c:/java-quick-start-master/src/main/java/com/mongodb/quickstart#
```

c.

```
GNU nano 3.2 ReadCDC.java
package com.mongodb.quickstart;

import com.mongodb.client.*;
import org.bson.Document;

import java.util.ArrayList;
import java.util.List;
import java.util.function.Consumer;

import static com.mongodb.client.model.Filters.*;
import static com.mongodb.client.model.Projections.*;
import static com.mongodb.client.model.Sorts.descending;

public class ReadCDC {

    public static void main(String[] args) {
        try (MongoClient mongoClient = MongoClient.create(System.getProperty("mongodb.uri"))) {
            MongoDBDatabase sampleTrainingDB = mongoClient.getDatabase("myDatabase");
            MongoCollection<Document> myCDCCollection = sampleTrainingDB.getCollection("myCollection");

            Document cdcDocument = myCDCCollection.find(new Document("recordId", "CDC")).first();
            System.out.println("CDC Record: " + cdcDocument.toJson());

        }
    }
}
```

d.

```
docker exec -it 4f38d2f55f7cb8e88b5fb15b28b2f41ac318e801b537bfc39f4812baf9d9b7d5 /bin/sh
root@4f38d2f55f7c:/java-quick-start-master/src/main/java/com/mongodb/quickstart# cd -
/java-quick-start-master
root@4f38d2f55f7c:/java-quick-start-master# mvn compile exec:java -Dexec.mainClass="com.mongodb.quickstart.ReadCDC" -Dmongodb.uri="mongodb://some-mongo:27017"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.mongodb:java-quick-start >-----
[INFO] Building java-quick-start 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ java-quick-start ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ java-quick-start ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 15 source files to /java-quick-start-master/target/classes
[INFO]
[INFO] --- exec-maven-plugin:3.0.0:java (default-cli) @ java-quick-start ---
CDC Record: {"_id": {"$oid": "61a9e99190675123a1a6e37b"}, "recordId": "CDC", "value": "Struct{after=Struct{record_num=1, route_number=1,id=y1894,bearing=90,current_status=IN_TRANSIT_TO,current_stop_sequence=1,direction_id=0,label=1894,latitude=42.34116000,longitude=-71.05764000,updated_at=1638523167000},source=Struct{version=1.5.2.Final,connector=mysql,name=localhost_MBTADB,ts_ms=1638525329633,snapshot=true,db=MBTADB,table=mbta_buses,server_id=0,file=binlog.000002,pos=84062,row=0},op=r,ts_ms=1638525329635}"}
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.787 s
[INFO] Finished at: 2021-12-03T11:00:18Z
[INFO]
root@4f38d2f55f7c:/java-quick-start-master#
```